SVM Tutorial: Min–Max, the Dual and the Kernel Trick
SOLUTIONS

Work individually. Ten questions, each building on the last. Write your working in the boxes.

Part A. Who moves first?

A parent chooses the *shop* and wants a low price; a kid chooses the *toy* and wants a high price.

Q1. Prices (Rs.):

	Toy 1	Toy 2
Shop A	50	80
Shop B	70	60

(a) Parent first: compute $\min_s \max_t p(s,t)$. (b) Kid first: compute $\max_t \min_s p(s,t)$. (c) Which is smaller, and why?

Solution. (a) Shop A → kid takes 80; Shop B → kid takes 70; parent picks B: **70**. (b) Toy 1 → parent finds 50; Toy 2 → parent finds 60; kid picks Toy 2: **60**. (c) Max–min = 60 < 70 = min–max. The inner player reacts to the outer choice, and reacting is an advantage.

Q2. Prices (Rs.):

	Toy 1	Toy 2	Toy 3
Shop A	3	5	4
Shop B	6	7	8
Shop C	2	9	1

Compute min–max and max–min. Find the entry that is the *largest in its row* and the *smallest in its column.*

Solution. Row maxima $5, 8, 9$, so min–max = 5. Column minima $2, 5, 1$, so max–min = 5. They are equal. The entry is **(Shop A, Toy 2)** = 5: the largest in row A and the smallest in column Toy 2. This is a **saddle point**.

Facts we will use

- **Weak duality:** for any table, $\max\min \le \min\max$.
- **Strong duality:** at a saddle point, $\max\min = \min\max$. The SVM problem always has one.

Part B. From the story to the SVM

For data $(\mathbf{x}_i, y_i)$, $y_i \in \{-1, +1\}$, the SVM "bill" is $L(\mathbf{w}, b, \alpha) = \frac{1}{2}\|\mathbf{w}\|^2 - \sum_i \alpha_i \big[y_i(\mathbf{w}\cdot\mathbf{x}_i + b) - 1\big]$, with $\alpha_i \ge 0$.

Q3. Complete the table.

	Variables	Wants L	Name / problem
Parent	$\mathbf{w}, b$	low	primal variables
Kid	$\alpha_1, \ldots, \alpha_n$	high	dual variables (Lagrange multipliers)
Primal	$\min_{\mathbf{w},b}\ \max_{\alpha\ge 0}\ L$	(parent commits first)	
Dual	$\max_{\alpha\ge 0}\ \min_{\mathbf{w},b}\ L$	(kid commits first)	

Q4. Take $\mathbf{w} = (1, 1)$, $b = -3$. Compute $g_i = y_i(\mathbf{w}\cdot\mathbf{x}_i + b) - 1$ and say what the kid does with α_i.

i	$\mathbf{x}_i$	y_i	$\mathbf{w}\cdot\mathbf{x}_i + b$	g_i	Kid's action
1	$(3, 2)$	$+1$	2	1	$\alpha_1 = 0$ (safe, ignored)
2	$(2, 2)$	$+1$	1	0	$\alpha_2 \ge 0$ allowed (on the margin: a support vector)
3	$(2, 0)$	$+1$	-1	-2	$\alpha_3 \to \infty$ (violation: this line is not allowed)
4	$(0, 1)$	-1	-2	1	$\alpha_4 = 0$ (safe, ignored)

Facts we will use

Solving the parent's reply and substituting back gives the **dual problem** and the **classifier**:

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i\alpha_j\, y_i y_j\, K(\mathbf{x}_i, \mathbf{x}_j) \quad \text{s.t. } \alpha_i \ge 0,\ \sum_i \alpha_i y_i = 0; \qquad f(\mathbf{x}) = \sum_i \alpha_i y_i\, K(\mathbf{x}_i, \mathbf{x}) + b.$$

With $K(\mathbf{x},\mathbf{z}) = \mathbf{x}\cdot\mathbf{z}$ this is the ordinary (linear) SVM, and then $\mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$. The **kernel trick**: replace K by any other kernel and nothing else changes. Support vectors satisfy $y_i f(\mathbf{x}_i) = 1$. In the **Gaussian (RBF) kernel** $K(\mathbf{x},\mathbf{z}) = e^{-\gamma\|\mathbf{x}-\mathbf{z}\|^2}$, the constant $\gamma > 0$ is a **hyperparameter**: chosen by the user, not learned. It sets how quickly similarity decays with distance (similarity drops to $e^{-1} \approx 0.37$ at distance $1/\sqrt{\gamma}$).

Part C. The kernel in practice

Definition (Gram matrix). Given points $\mathbf{x}_1, \ldots, \mathbf{x}_n$ and a kernel K, the **Gram matrix** is the $n \times n$ table of all pairwise kernel values,

$$\mathbf{K} = \begin{pmatrix} K(\mathbf{x}_1,\mathbf{x}_1) & \cdots & K(\mathbf{x}_1,\mathbf{x}_n) \\ \vdots & \ddots & \vdots \\ K(\mathbf{x}_n,\mathbf{x}_1) & \cdots & K(\mathbf{x}_n,\mathbf{x}_n) \end{pmatrix}, \qquad K_{ij} = K(\mathbf{x}_i,\mathbf{x}_j).$$

Entry K_{ij} measures how similar points i and j are. It is all the dual problem needs from the data: training an SVM means working with this one matrix.

Q5. A complete SVM by hand (linear kernel). Points $\mathbf{x}_1 = (1,1)$, $y_1 = +1$ and $\mathbf{x}_2 = (-1,-1)$, $y_2 = -1$. (a) Use $\sum_i \alpha_i y_i = 0$ to show $\alpha_1 = \alpha_2 = \alpha$. (b) Write the dual objective as a function of α and maximise it. (c) Find $\mathbf{w}$ and b, and write $f(\mathbf{x})$.

Solution. (a) $\alpha_1 - \alpha_2 = 0$. (b) $K_{11} = K_{22} = 2$, $K_{12} = -2$; every $y_i y_j K_{ij} = 2$, so the double sum is $8\alpha^2$ and the objective is $2\alpha - 4\alpha^2$. Setting the derivative $2 - 8\alpha = 0$ gives $\alpha = \frac{1}{4}$. (c) $\mathbf{w} = \frac{1}{4}(1,1) - \frac{1}{4}(-1,-1) = (\frac{1}{2}, \frac{1}{2})$. From $y_1 f(\mathbf{x}_1) = 1$: $1 + b = 1$, so $b = 0$. $f(\mathbf{x}) = \frac{1}{2}x_1 + \frac{1}{2}x_2$: the line $x_1 + x_2 = 0$, exactly halfway between the points.

Q6. Gram matrices. Using the definition above, for $\mathbf{x}_1 = (1,0)$, $\mathbf{x}_2 = (0,1)$, $\mathbf{x}_3 = (1,1)$, write the 3×3 matrix $K_{ij} = K(\mathbf{x}_i,\mathbf{x}_j)$ for (a) $K(\mathbf{x},\mathbf{z}) = \mathbf{x}\cdot\mathbf{z}$ and (b) $K(\mathbf{x},\mathbf{z}) = (\mathbf{x}\cdot\mathbf{z})^2$. What is K_{23} in each case?

Solution.

(a) $K = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ (b) $K = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 4 \end{pmatrix}$ (square each entry of (a))

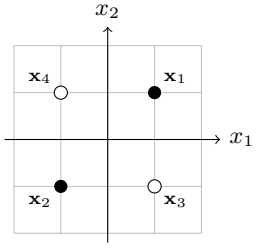
$K_{23} = 1$ in both. Both matrices are symmetric: $K_{ij} = K_{ji}$.

Q7. The trick itself. For $K(\mathbf{x},\mathbf{z}) = (\mathbf{x}\cdot\mathbf{z})^2$ the hidden features are $\phi(\mathbf{x}) = (x_1^2,\ \sqrt{2}\,x_1x_2,\ x_2^2)$. For $\mathbf{x} = (2,1)$, $\mathbf{z} = (1,3)$ compute $K(\mathbf{x},\mathbf{z})$ directly, then compute $\phi(\mathbf{x})\cdot\phi(\mathbf{z})$. What do you notice?

Solution. Directly: $\mathbf{x}\cdot\mathbf{z} = 5$, so $K = 25$. Via features: $\phi(\mathbf{x}) = (4, 2\sqrt{2}, 1)$, $\phi(\mathbf{z}) = (1, 3\sqrt{2}, 9)$, dot product $4 + 12 + 9 = 25$. Same answer: the kernel computes a dot product in feature space *without building the features.*

Q8. Kernel trick in action: XOR. $\mathbf{x}_1 = (1,1)$, $\mathbf{x}_2 = (-1,-1)$ have $y = +1$ (filled); $\mathbf{x}_3 = (1,-1)$, $\mathbf{x}_4 = (-1,1)$ have $y = -1$ (hollow). Use $K(\mathbf{x},\mathbf{z}) = (\mathbf{x}\cdot\mathbf{z})^2$.

(a) Look at the figure. Can any straight line separate the two classes?
(b) Compute the 4×4 Gram matrix.
(c) Take $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = a$. Check $\sum \alpha_i y_i = 0$, write the dual objective in terms of a, and maximise.
(d) Using $\mathbf{x}_1$, find b. Then write $f(\mathbf{x}) = \sum_i \alpha_i y_i K(\mathbf{x}_i,\mathbf{x}) + b$ and simplify it in terms of x_1, x_2.
(e) Classify $(2,3)$ and $(-1,2)$ using *only kernel evaluations*, and check against (d).



Solution. (a) No: the $+1$ points sit on one diagonal and the -1 points on the other.

(b) $K = \begin{pmatrix} 4 & 4 & 0 & 0 \\ 4 & 4 & 0 & 0 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 4 & 4 \end{pmatrix}$ ($\mathbf{x}_i\cdot\mathbf{x}_j = \pm 2$ within a class, 0 across classes).

(c) $a + a - a - a = 0$ ✓. $y_i y_j K_{ij} = 4$ for the 8 within-class pairs, else 0; the objective is $4a - \frac{1}{2}\cdot 32a^2 = 4a - 16a^2$, so $4 - 32a = 0$ gives $a = \frac{1}{8}$.

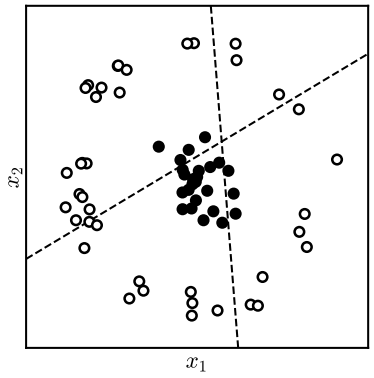
(d) $f(\mathbf{x}_1) = \frac{1}{8}(4 + 4 - 0 - 0) + b = 1 + b = 1 \Rightarrow b = 0$. $f(\mathbf{x}) = \frac{1}{8}\big[2(x_1 + x_2)^2 - 2(x_1 - x_2)^2\big] = \boxed{x_1x_2}$: positive in quadrants I and III, negative in II and IV.

(e) $(2,3)$: kernels $25, 25, 1, 1$, so $f = \frac{1}{8}(25 + 25 - 1 - 1) = 6 > 0$: class $+1$ ($= 2\cdot 3$ ✓). $(-1,2)$: kernels $1, 1, 9, 9$, so $f = \frac{1}{8}(1 + 1 - 9 - 9) = -2 < 0$: class -1 ($= -1\cdot 2$ ✓).

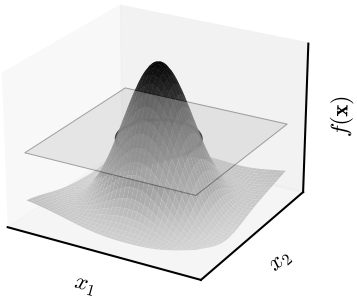
Insight: in feature space the middle coordinate $\sqrt{2}\,x_1x_2$ is $+\sqrt{2}$ for one class and $-\sqrt{2}$ for the other, so a flat cut separates them. We never built ϕ; the kernel did it.

Illustration: the Gaussian kernel before and after.

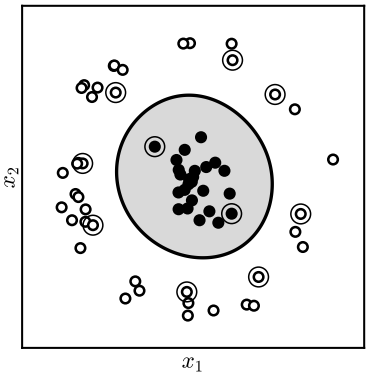
(a) Before: no straight line works



(b) Kernel trick: $f(\mathbf{x})$ as a landscape, flat plane $f = 0$ is sea level



(c) After: boundary $f(\mathbf{x}) = 0$



(a) *Data.* 25 points with $y_i = +1$ (filled) satisfy $\|\mathbf{x}_i\| \le 0.8$; 40 points with $y_i = -1$ (hollow) satisfy $1.6 \le \|\mathbf{x}_i\| \le 2.1$. No $(\mathbf{w}, b)$ achieves $y_i(\mathbf{w} \cdot \mathbf{x}_i + b) > 0$ for all i, so a linear SVM fails.

(b) *Kernel.* With $K(\mathbf{x}, \mathbf{z}) = e^{-0.5\|\mathbf{x}-\mathbf{z}\|^2}$ the trained SVM is $f(\mathbf{x}) = \sum_{i \in \mathrm{SV}} \alpha_i y_i \, e^{-0.5\|\mathbf{x}_i - \mathbf{x}\|^2} + b$. Each term with $y_i = +1$ is a Gaussian bump centred at $\mathbf{x}_i$; each term with $y_i = -1$ is a Gaussian dip. The surface shows $f(\mathbf{x})$ as height; the flat plane is $f = 0$.

(c) *Result.* The classifier is $\hat{y} = \operatorname{sign} f(\mathbf{x})$, and the boundary $\{\mathbf{x} : f(\mathbf{x}) = 0\}$ is a closed curve around the $+1$ points. Only 10 of the 65 points have $\alpha_i > 0$ (circled). The feature map ϕ is never formed: f needs only the values $K(\mathbf{x}_i, \mathbf{x})$.

Q9. RBF kernel as similarity. For $K(\mathbf{x}, \mathbf{z}) = e^{-\gamma\|\mathbf{x}-\mathbf{z}\|^2}$, $\mathbf{x} = (0, 0)$, $\mathbf{z} = (1, 1)$, compute K for $\gamma = 0.5$ and $\gamma = 5$. (Use $e^{-1} \approx 0.37$, $e^{-10} \approx 0.00005$.) Which γ says "these points are similar"?

Solution. $\|\mathbf{x} - \mathbf{z}\|^2 = 2$. $\gamma = 0.5$: $K = e^{-1} \approx 0.37$ (fairly similar). $\gamma = 5$: $K = e^{-10} \approx 0.00005$ (unrelated). Small γ: each support vector's influence reaches far (smooth boundary). Large γ: influence is very local (wiggly boundary, risk of overfitting).

Q10. Predicting with an RBF SVM. A trained SVM ($\gamma = 1$, $b = 0$) has two support vectors: $\mathbf{x}_1 = (0, 0)$, $y_1 = +1$, $\alpha_1 = 1$ and $\mathbf{x}_2 = (2, 0)$, $y_2 = -1$, $\alpha_2 = 1$. So $f(\mathbf{x}) = K(\mathbf{x}_1, \mathbf{x}) - K(\mathbf{x}_2, \mathbf{x})$. (a) Classify $(0.5, 0)$ and $(1.5, 0)$. (Use $e^{-0.25} \approx 0.78$, $e^{-2.25} \approx 0.11$.) (b) What is $f(1, 0)$? (c) Describe the decision boundary.

Solution. (a) $(0.5, 0)$: $f = e^{-0.25} - e^{-2.25} \approx 0.78 - 0.11 = 0.67 > 0$: class $+1$. $(1.5, 0)$: $f = e^{-2.25} - e^{-0.25} \approx -0.67 < 0$: class -1. (b) Equidistant from both, so $f = e^{-1} - e^{-1} = 0$: on the boundary. (c) $f = 0$ exactly where a point is equally similar to both support vectors, i.e. equidistant: the vertical line $x_1 = 1$. The prediction is a *similarity-weighted vote* of the support vectors.